

AI Support Ops Assistant

Skill & Agent Design Package

"A capability-by-capability assessment of where AI adds value, where deterministic automation is sufficient, and where neither should be introduced."

Submitted by **Nisha Verma** · 25 August 2026

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SECTION 01

Business Objective & Design Approach

BUSINESS CONTEXT

A mid-size SaaS company operates support through one shared inbox and a ticketing tool. Five support activities are currently performed manually and inconsistently:

- Ticket categorization and routing
- Customer reply drafting
- At-risk customer briefing
- Weekly support-quality review
- Bug investigation and engineering write-up

BUSINESS OBJECTIVE

Leadership wants to improve support operations by reducing support handling time, improving consistency, improving policy compliance, reducing manual preparation work, preventing incorrect customer communication, improving engineering handoffs — and avoiding unnecessary AI complexity.

"AI is introduced only where it provides meaningful value beyond deterministic automation."

FOUR POSSIBLE DECISIONS

Deterministic Known rules + predictable flow.	Skill Bounded AI interpretation, synthesis, classification, or generation inside a predictable procedure.	Agent Dynamic investigation + model-directed tool/action selection.	Neither AI does not provide enough reliable value to justify automation.
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"Choose the least autonomous abstraction that reliably solves the business problem."

Five Capabilities. Five Different Decisions.

The classification is based on the nature of the work, not on how sophisticated the technology sounds.

CAPABILITY	BUSINESS OBJECTIVE	DECISION	WHY	AI VALUE VS DETERMINISTIC
Ticket Triage	Route incoming tickets quickly and consistently while immediately escalating outages, legal, and security issues.	Skill	Bounded triage decision using a predefined taxonomy, routing rules, priorities, and escalation rules. AI interprets natural-language descriptions but does not need autonomous investigation.	Handles natural-language variation while deterministic rules remain authoritative for routing and escalation.
Reply Drafting	Increase response speed and consistency while ensuring customer replies follow current policy and company tone.	Skill	Bounded generation using current policy, tone guidance, and customer context.	Produces contextual responses beyond rigid templates while policy remains the source of truth.
At-Risk Customer Briefing	Reduce retention-call preparation time and create a consistent evidence-based customer-health briefing.	Skill	The same sources are required each time and structured exceptions can be deterministic. AI's role is synthesis.	Turns heterogeneous records into a concise narrative and surfaces signals/contradictions.
Weekly QA Grading	Identify recurring support-quality problems.	Neither	Subjective judgments lack sufficiently objective ground truth. Incorrect results could misdirect coaching and operational decisions.	Not sufficiently reliable to justify automation within this scope.
Bug Triage	Reduce duplicate engineering work and shorten the path from customer-reported bug to actionable engineering issue.	Agent	Requires knowledge retrieval, semantic interpretation, decision-making, and conditional action.	Dynamic semantic investigation and model-directed conditional action.

Ticket Triage

"Route every incoming support request to the correct destination quickly while ensuring outages, legal issues, and security concerns receive immediate escalation."

CURRENT OPERATIONAL PROBLEM

- Manual categorization is inconsistent.
- Routing depends on individual interpretation.
- High-risk tickets can be delayed.
- Support volume makes consistent classification difficult.

WHY SKILL?

The output is a bounded triage decision. Categories, teams, priorities, and escalation boundaries are predefined. AI is useful because customers describe the same problem in different natural-language forms — this needs bounded interpretation, not autonomous investigation.

WHY NOT AGENT?

An Agent would introduce unnecessary autonomy. There is no need for iterative investigation, open-ended tool selection, or model-directed actions.

BUSINESS VALUE VS DETERMINISTIC AUTOMATION

AI adds value by interpreting varied customer language and mapping it to a controlled taxonomy. Deterministic rules remain the source of truth for routing, priority, escalation, and team assignment.

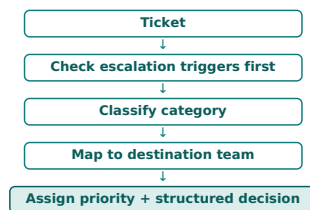
THE SKILL DOES

- Classifies tickets
- Routes to the correct team
- Assigns priority
- Flags mandatory escalation

CAPABILITY BOUNDARY

- Investigate customer history
- Reply to customers
- Fix issues
- Decide outside the defined rulebook

WHAT'S INSIDE THIS SKILL



ESCALATION TRIGGERS

Live outage: "down," "500/503 error," widespread access failure, "everything broken."
Legal/Security: "breach," "hacked," "GDPR/CCPA," "subpoena," "PII exposed." Either triggers immediate P0 escalation; if both plausibly apply, Legal/Security wins as the category.

PRIORITY LEVELS

P0 — reserved for escalations only. **P1** — blocks one paying customer's core workflow. **P2** — real issue, workaround exists. **P3** — no urgency (questions, feature requests).

Category	→ Destination Team
Outage	→ Engineering
Legal/Security	→ Legal/Security Team
Technical Issue	→ Engineering
Billing	→ Billing Team
Account/Access	→ Support L1
Feature Request	→ Product
General Inquiry	→ Support L1

WHY THIS LOGIC MATTERS

Escalation is checked before anything else — a ticket matching a live-outage or legal/security signal is escalated regardless of what other category it might also fit, so an urgent issue can never be buried under a lower-priority label.

PASS

Live outage

PASS

Technical issue

PASS

Security issue

PASS

Billing issue

IMPLEMENTATION SCOPE

The current implementation is optimized for the categories and patterns defined in the operational rulebook. Broader semantic generalization would be a natural next iteration as the taxonomy and examples expand.

Reply Drafting

"Increase response speed while ensuring every customer-facing reply matches company tone and never states unsupported refund or pricing commitments."

CURRENT PROBLEM

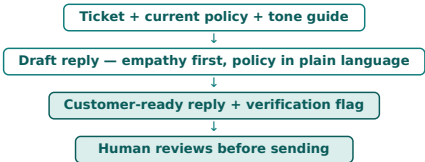
- Agents manually interpret policy.
- Policy changes frequently.
- Tone varies between agents.
- Incorrect policy statements create customer/business risk.

WHY SKILL? Bounded generation. Inputs are customer context + current policy + tone guidance. Output is a customer-ready draft. No autonomous investigation or action selection is required.	WHY NOT AGENT? The procedure is predictable. There is no need for the model to dynamically choose tools or investigate multiple systems.	WHY NOT PURELY DETERMINISTIC? Rigid templates handle common cases but become brittle when requests vary or multiple policy constraints must be combined.
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BUSINESS VALUE VS DETERMINISTIC AUTOMATION
AI provides contextual language generation. Deterministic policy remains the source of truth — the AI turns approved information into natural customer-facing language.

THE SKILL DOES • Drafts replies • Follows company tone • Uses current policy • Flags verification where necessary	CAPABILITY BOUNDARY • Approve refunds • Invent policy • Send replies • Make business commitments
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WHAT'S INSIDE THIS SKILL



REFUND POLICY Full refund within 14 days (once verified). Up to 50% discretionary refund 15-30 days (never guaranteed). Duplicate charges refunded once a system error is confirmed (~5-7 day estimate). No refunds after 30 days except confirmed billing errors.	tone principles Empathy first — acknowledge the situation before citing policy. Avoid jargon and over-apologizing. 3-5 sentences, resolution-led. Sign off with "Best," or "Thanks," — never "Sincerely."
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Pricing (published, safe to state directly)
Basic: \$9/mo Pro: \$29/mo Team: \$79/mo
Annual billing: 15% discount
No custom discounts beyond published promotions

WHY THIS LOGIC MATTERS
The reply never confirms that a specific customer qualifies — it explains what the policy generally allows, then routes anything customer-specific (a refund, a discount, an exception) to human verification, even when the case looks clear-cut.

TEST EVIDENCE

15/15 Test cases run	13 Verification required	2 No verification needed	0 Unsupported settled outcomes
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At-Risk Customer Briefing

"Reduce preparation time for retention calls and provide account teams with a consistent, evidence-based view of customer health."

CURRENT PROBLEM

Relevant information is fragmented across **Billing + Product Usage + Past Tickets**. Humans manually consolidate the information and may miss contradictions or meaningful signals.

WHY SKILL?

The retrieval pattern is predictable. The same sources are required each time. Structured exceptions can be handled deterministically. The AI-specific problem is synthesis.

WHY NOT AGENT?

Open-ended investigation is unnecessary. The data sources and retrieval pattern are already known.

WHY NOT PURELY DETERMINISTIC?

Deterministic automation can retrieve and aggregate records. It is less suited to turning heterogeneous evidence into a concise narrative for a human retention conversation.

BUSINESS VALUE VS DETERMINISTIC AUTOMATION

AI adds value by synthesizing heterogeneous evidence, separating facts from interpretation, surfacing contradictions, organizing meaningful signals, and producing a standardized one-page briefing.

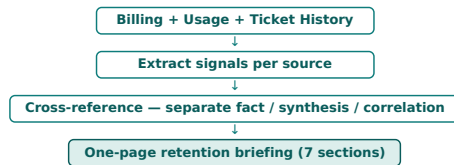
THE SKILL DOES

- Synthesizes retrieved information
- Separates fact / synthesis / correlation
- Surfaces unresolved issues & gaps

CAPABILITY BOUNDARY

- Independently investigate new sources
- Diagnose root causes
- Contact the customer
- Make retention decisions

WHAT'S INSIDE THIS SKILL



SIGNAL VS. NOISE

Billing: late/failed payments, disputes, downgrades — not routine on-time history. **Usage:** sustained decline, feature abandonment — not normal fluctuation. **Tickets:** repeated, escalated, or unresolved issues — not routine how-to requests.

BRIEFING STRUCTURE

Customer → Overall Risk Picture → Why the Customer May Be at Risk → Customer History That Matters → Key Unresolved Issues → What to Know Before the Call → Data Gaps (only when something is genuinely missing or conflicting).

THE CORE RULE: CORRELATION VS. CAUSATION

Causal language ("caused," "root cause," "drove") is only used when the source data explicitly supports that specific link. Two signals occurring in the same period default to correlation phrasing — "coincided with," never "caused by." This is the rule that Test 1 (below) exposed a real gap in, and fixed.

VALIDATION LEARNING

OBSERVED

Billing lateness and usage decline were being combined into one unsupported root-cause claim.

DESIGN RESPONSE

Introduced an explicit correlation-vs-causation rule into synthesis-guide.md.

RE-VALIDATED

Re-ran the test suite against the corrected rule; capability re-verified.

5/5 PASS

Live scenarios

Weekly QA Grading

"Identify recurring support-quality problems and improve coaching, process quality, and customer experience."

WHY NEITHER?

The capability requires judging human support interactions against a rubric and identifying recurring failure patterns. The underlying judgments do not have sufficiently objective ground truth. Incorrect judgments could influence coaching and operational decisions.

WHY NOT SKILL?

A Skill could produce plausible scores, but plausibility is not enough when outputs may affect employee coaching and operational decisions.

WHY NOT AGENT?

More autonomy does not solve the fundamental reliability problem. The task also requires recurring weekly sampling/orchestration, which is outside this package.

WHY NOT PURELY DETERMINISTIC?

Explicit rubric checks can be automated, but qualitative judgments and recurring patterns cannot safely be reduced to fixed rules without losing the purpose of the review.

BUSINESS VALUE / BENEFIT

Potential benefit exists, but reliability risk is too high in the current scope.

NOT BUILT.

"Correctly choosing not to automate is part of the design decision."

"The goal is not maximum AI coverage. The goal is appropriate AI coverage."

Bug Triage

"Reduce duplicate engineering work and shorten the path from customer-reported bug to actionable engineering issue."

CURRENT PROBLEM

- Support manually searches the changelog.
- Customer language differs from engineering terminology.
- Known issues can be duplicated.
- New issues require manual write-ups.
- Correct action depends on investigation results.

WHY AGENT?

The model must inspect the knowledge source, interpret search results, determine whether a candidate is a genuine match, decide what to do based on evidence, and produce a structured write-up when no sufficiently matching issue exists.

WHY NOT SKILL?

This is not simply input → generation → output. The model-directed control flow changes depending on intermediate evidence.

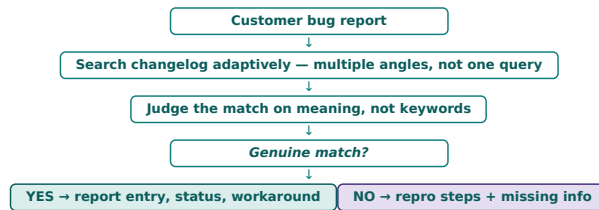
WHY NOT DETERMINISTIC?

Literal/keyword lookup produces false positives and negatives because customer and engineering language differ.

BUSINESS VALUE VS DETERMINISTIC AUTOMATION

The Agent provides semantic interpretation, adaptive investigation, candidate evaluation, and conditional action.

WHAT'S INSIDE THIS AGENT



TOOL BOUNDARY

[Read](#)[Grep](#)[Glob](#)

Read-only access to the changelog and related files — the agent cannot write, edit, or run arbitrary commands. It never creates a ticket itself; a "new_issue" result is a structured write-up handed to a human or downstream system, not an automated action.

CAPABILITY BOUNDARY

- Reply to customers
- Route tickets
- Determine severity
- Inspect customer history
- Check billing
- Fix bugs
- Update the changelog

GROUNDING RULE

Engineering issue details and reproduction steps must come from the customer report. The Agent must not invent facts, environment details, error messages, or reproduction steps.

Known issue — Priya's ticket

Matched to v4.12.3 on meaning alone (zero literal keyword overlap between "spins forever" and "fails to fully render... timeout"). Correctly ruled out two other plausible entries.

New issue — Marcus's ticket

Correctly avoided a false match on the one entry with surface-level keyword overlap ("report"). Produced repro steps + missing-info list without inventing details.

SECTION 09

Evidence & Final Decision

Ticket Triage — Skill

All four escalation/routing test cases passed.

4/4 PASS

Reply Drafting — Skill

13 verification-required · 2 no-verification · 0 unsupported settled outcomes.

15/15 PASS

At-Risk Briefing — Skill

Correlation-vs-causation issue found and fixed during testing.

5/5 PASS

Weekly QA — Neither

Rationale: insufficient reliable ground truth.

NOT BUILT

Bug Triage — Agent

Known issue matched purely on meaning, zero keyword overlap with the changelog entry. New issue correctly kept separate from a surface-level false match; repro steps extracted without inventing details.

2/2 PASS

FINAL IMPLEMENTATION SCOPE

"All built capabilities were tested independently." "No end-to-end orchestration, coordinator, or full pipeline was built."

"Choose the least autonomous abstraction that reliably solves the business problem."

Deterministic where possible →

Skill where bounded AI judgment adds value →

Agent where dynamic investigation & conditional action are required →

Neither where AI is not reliably justified